1-W1　两个同号固定电荷之间的平衡点

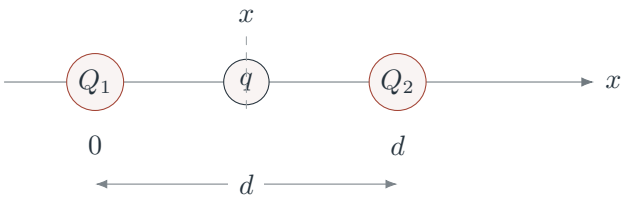
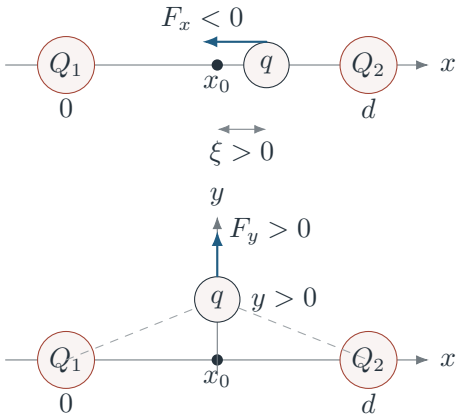


图 1-7　题图　同号源之间的连线约束

$q>0$；ξ, y 均为相对平衡点的微扰

轴向：$F_x \simeq -2kqA\xi$



横向：$F_y \simeq kqAy$，偏离被放大

图 1-8　解答分析图　正试探电荷：轴向小位移产生恢复力，横向小位移产生发散力。

1-W2　异号电荷的平衡位置

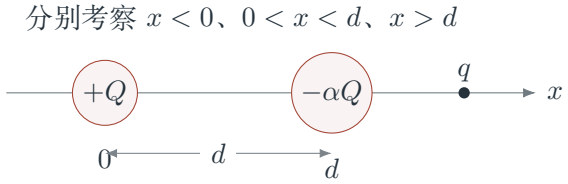
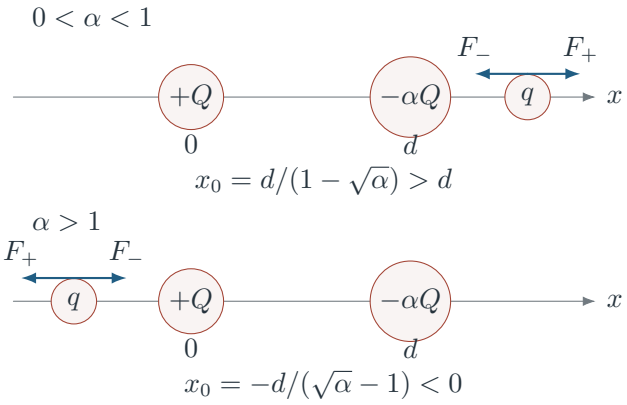


图 1-9　题图　异号源与外侧场点



箭头取 $q>0$；$\alpha=1$ 时无有限平衡点

图 1-10　解答分析图　图示右侧候选点；需按电量分区

1-W3　悬挂带电小球：精确方程与小角近似

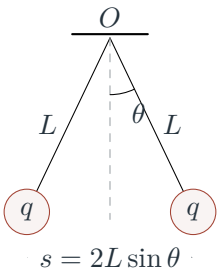


图 1-11　题图　悬挂双球的几何

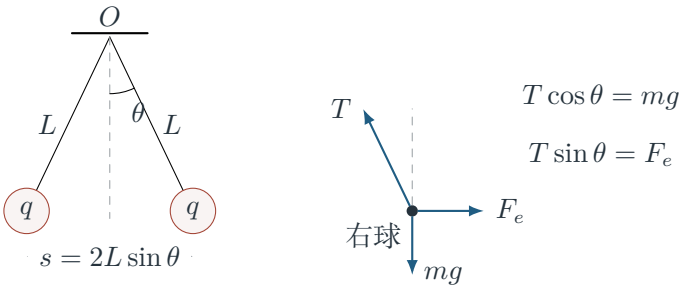


图 1-12　解答分析图　右球的三个力与分量平衡

1-W4　三电荷体系的平衡

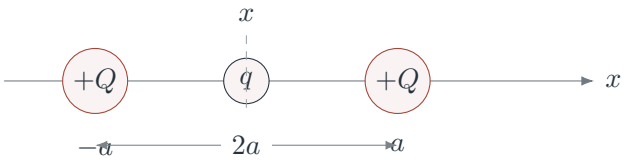
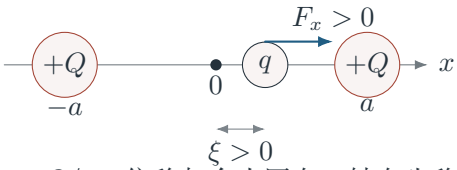


图 1-13　题图　三个电荷的平衡构型；两端为 $\pm a$

$$F_x \simeq kQ^2\xi/a^3$$



$q=-Q/4$：位移与合力同向，轴向失稳

图示势能变分：两端位置暂保持不变

图 1-14　解答分析图　中心负电荷的轴向微扰使合力与位移同向；对应总势能的下降方向。

1-W5　等边三角形中的合力与补偿电荷

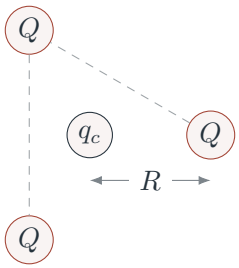


图 1-15　题图　等边三角形与中心电荷

两条邻顶点斥力投影相加；中心负电荷向内补偿

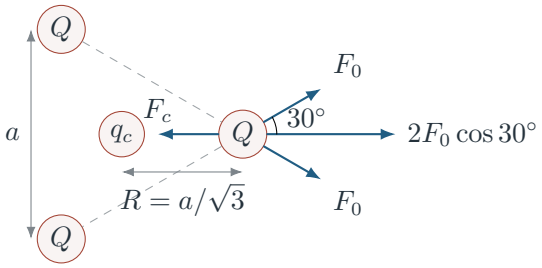
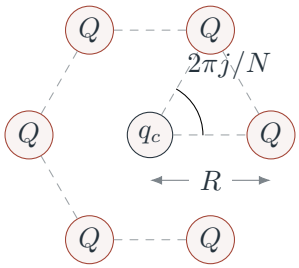


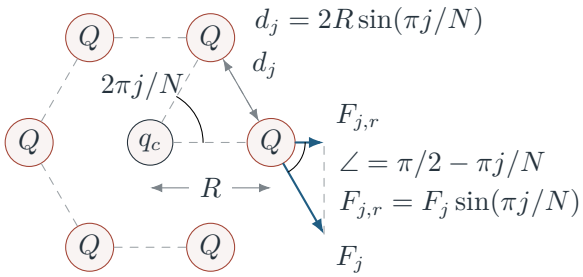
图 1-16　解答分析图　顶点合力方向沿中心至顶点向外

1-W6　正多边形顶点电荷的合力



绘图以 $N = 6$ 示意；公式适用于一般 N

图 1-17　题图　正多边形的代表性几何



绘图以 $N = 6$ 示意；公式适用于一般 N

图 1-18　解答分析图　弦长与径向投影决定求和项

1-W7　库仑力与弹簧：平衡、频率及失稳阈值

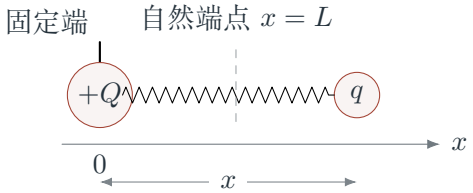


图 1-19　题图　带电粒子与弹簧的轴线模型

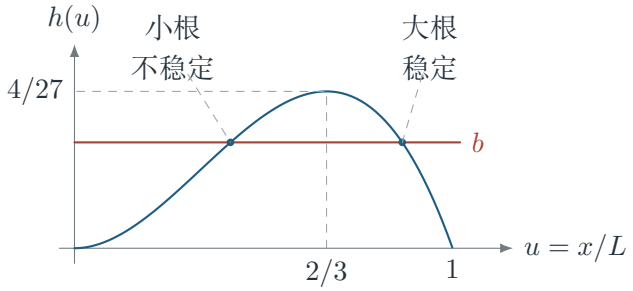


图 1-20　解答分析图　异号分支的平衡根、临界值与稳定性

1-W8　从力的比值反推距离指数

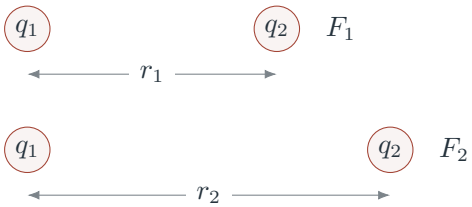
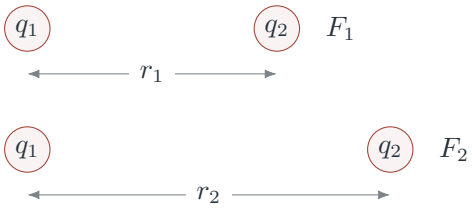


图 1-21　题图　两次测力的距离与电荷条件

$$F_1/F_2 = (r_2/r_1)^{2+\delta}$$



比值消去未知电荷乘积与比例常数

图 1-22　解答分析图　用比值消去未知常量

1-T1　二维力的叠加

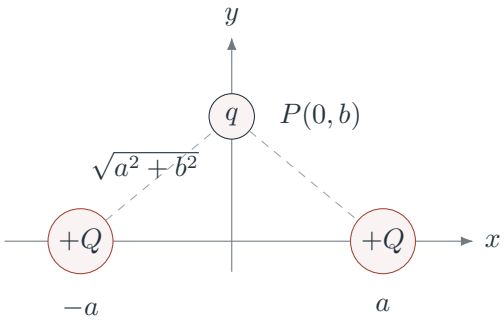


图 1-23　题图　两源与二维场点

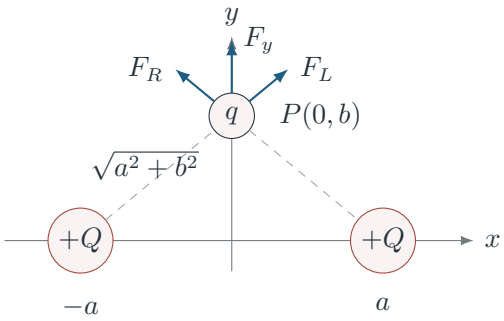


图 1-35　解答分析图　图中取 $q>0$，水平方向抵消

1-T2　不等电荷之间的平衡

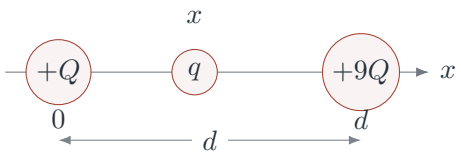


图 1-24　题图　不等正电荷间的约束粒子

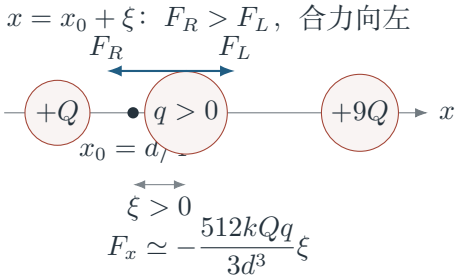


图 1-36　解答分析图　平衡点更靠近较弱源

1-T3　异号电荷的区域判断

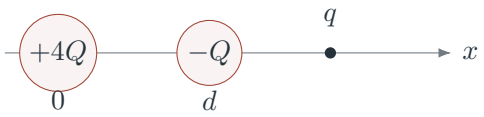


图 1-25　题图　右侧候选点；两源之间与左侧另行判断

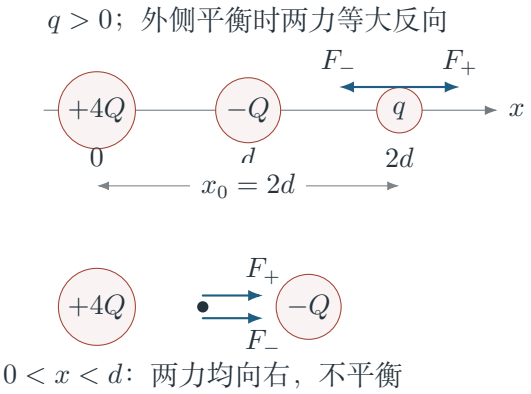


图 1-37　解答分析图　只有较弱源外侧可能平衡

1-T4　接触前后的力

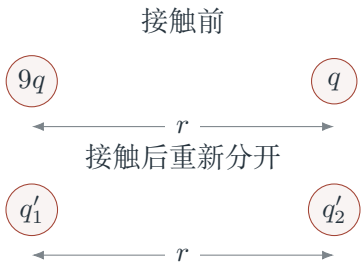


图 1-26　题图　相同小球接触前后

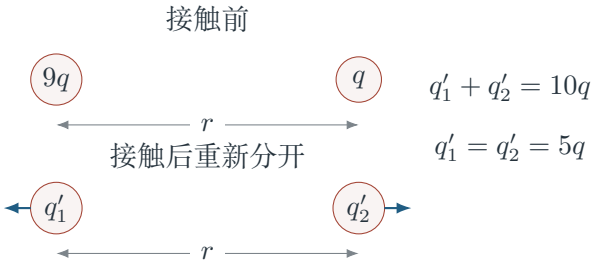


图 1-38　解答分析图　守恒与相同小球的对称分配

1-T5　正方形与中心电荷

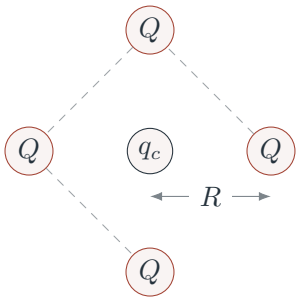
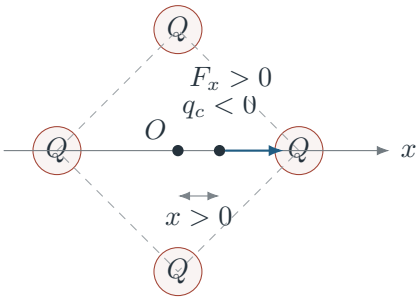


图 1-27　题图　正方形顶点与中心电荷；边长为 a



沿轴的两源：$E_x \simeq -4kQx/R^3$

垂直轴的两源：$E_x \simeq +2kQx/R^3$

$R = a/\sqrt{2}$；总场向左，负中心电荷受力向右

中心电荷的平面内位移被放大

图 1-39　解答分析图　中心负电荷偏移后，两对源的场相加向左，粒子受力向右而失稳。

1-T6　带电小球的比例律

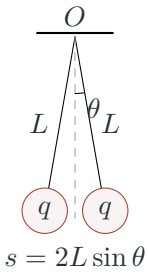
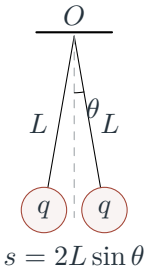
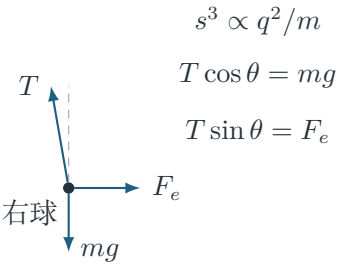


图 1-28　题图　悬挂双球的电荷与质量改变





$$s^3 \propto q^2/m$$

$$T\cos\theta = mg$$

$$T\sin\theta = F_e$$

图 1-40　解答分析图　受力和几何关系共同决定比例律

1-T7　无需小角近似的反演

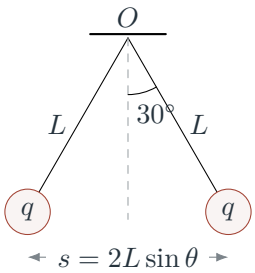


图 1-29　题图　有限偏角的悬挂双球

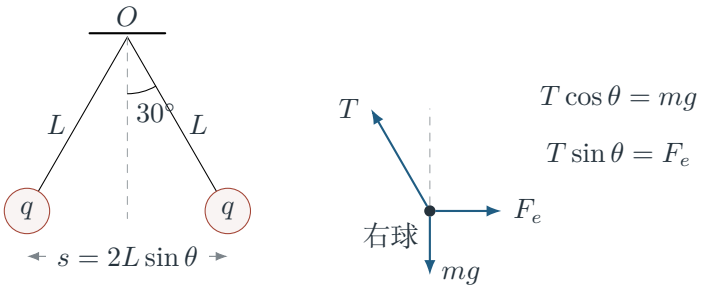


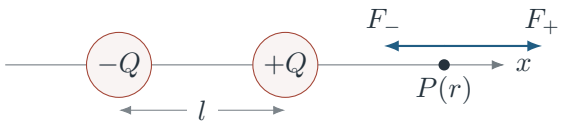
图 1-41　解答分析图　使用精确三角函数关系

1-T8　总电荷为零仍有作用力



图 1-30　题图　中性电荷对的外侧场点

$r \gg l$：两项相减后保留 r^{-3}



$q>0$：$F_+ > F_-$；$q<0$ 时全部反向

图 1-42　解答分析图　正负贡献相减后保留首个非零项

1-T9　挑战：稳定性取决于允许运动的方向

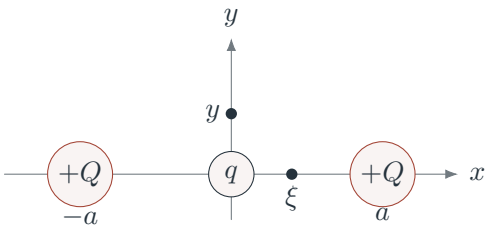
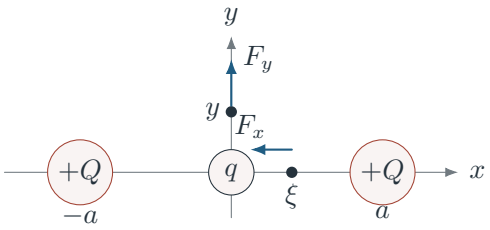


图 1-31　题图　轴向和横向的两种约束



$q>0$：轴向恢复，横向偏离被放大

图 1-43　解答分析图　同一平衡位置的两种一阶受力

1-T10　挑战：三个自由电荷的平衡条件

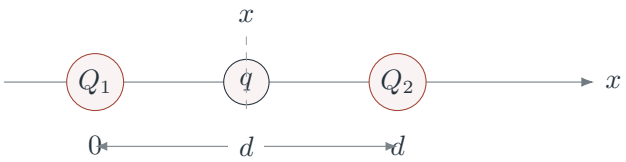


图 1-32　题图　三个自由电荷各自平衡

$q<0$；中心平衡与两端平衡须同时成立

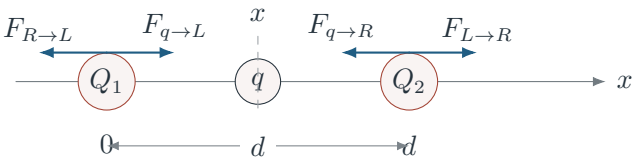


图 1-44　解答分析图　对两端列方程，再检查中心

1-T11　挑战：临界平衡的一阶系数为零

$$B = kQ|q| = 4\kappa L^3/27$$

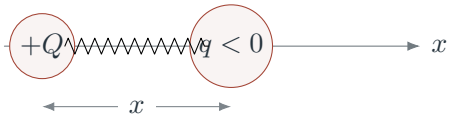
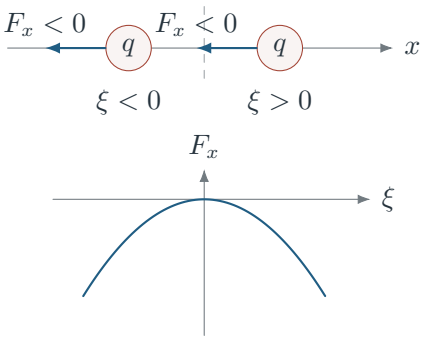


图 1-33　题图　异号弹簧模型的临界位置

$$x_0 = 2L/3$$



$$F_x \simeq -9\kappa\xi^2/(4L)$$

图 1-45　解答分析图　首个非零项是二次项，方向不随位移变号

1-T12　挑战：从两次测力反演电荷

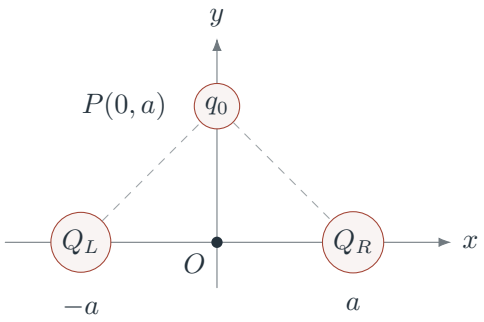
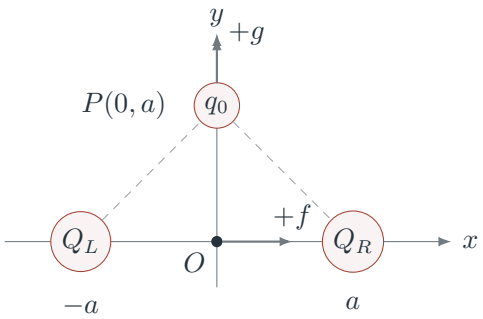


图 1-34　题图　两个测点与待反演电荷



箭头表示正分量参考方向；f, g 可正可负

$$f \propto Q_L - Q_R,\ \ g \propto Q_L + Q_R$$

图 1-46　解答分析图　箭头标示分量正方向；测得的 f, g 可正可负